Exercise - 18.3

① Find, from first principle the derivatives of

a) $y = \cos^{-1} x$

$\cos y = x$

differentiating with respect to x.

$\frac{d \cos y}{dx} = \frac{dx}{dx}$

or, $\frac{d \cos y}{dy} \times \frac{dy}{dx} = 1$

or, $-\sin y \times \frac{dy}{dx} = 1$

or, $\frac{dy}{dx} = -\frac{1}{\sin y}$

or, $\frac{dy}{dx} = \frac{-1}{\sqrt{1-\cos^2 y}}$

or, $\frac{dy}{dx} = \frac{-1}{\sqrt{1-x^2}}$

b) $y = \sin^{-1} 2x$

or, $\sin y = 2x$

differentiating with respect to x.

$\frac{d \sin y}{dx} = \frac{d 2x}{dx}$

or, $\frac{d \sin y}{dy} \times \frac{dy}{dx} = 2$

or, $\cos y \times \frac{dy}{dx} = 2$

or, $\frac{dy}{dx} = \frac{2}{\cos y}$

or, $\frac{dy}{dx} = \frac{2}{\sqrt{1-\sin^2 x}}$

$\frac{dy}{dx} = \frac{2}{\sqrt{1-(2x)^2}}$

$\frac{dy}{dx} = \frac{2}{\sqrt{1-4x^2}}$ #.

c) $y = \tan^{-1} ax$

or, $\tan y = ax$

differentiating with respect to x.

$\frac{d \tan y}{dx} = \frac{d ax}{dx}$

or, $\frac{d \tan y}{dy} \times \frac{dy}{dx} = a$

or, $\sec^2 y \times \frac{dy}{dx} = a$

or, $\frac{dy}{dx} = \frac{a}{\sec^2 y}$

$\frac{dy}{dx} = \frac{a}{1+\tan^2 y}$

$\frac{dy}{dx} = \frac{a}{1+(ax)^2}$ #.

(2) Find the derivative of following with respect to x.

a) $y = \sin^{-1} x + \cos^{-1} x$

differentiating with respect to x.

$\frac{dy}{dx} = \frac{d \sin^{-1} x}{dx} + \frac{d \cos^{-1} x}{dx}$

or, $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} + \left(-\frac{1}{\sqrt{1-x^2}}\right)$

$\therefore \frac{dy}{dx} = 0$ //.

b) $y = \tan^{-1}x + \cot^{-1}x$

differentiating with respect to x.

$\frac{dy}{dx} = \frac{d(\tan^{-1}x + \cot^{-1}x)}{dx}$

$= \frac{d\tan^{-1}x}{dx} + \frac{d\cot^{-1}x}{dx}$

$= \frac{1}{1+x^2} + \left(-\frac{1}{1+x^2}\right)$

$= 0$

$\therefore \frac{dy}{dx} = 0$ //.

c) $y = \cot^{-1}2x$

differentiating with respect to x.

$\frac{dy}{dx} = \frac{d\cot^{-1}2x}{d2x} \times \frac{d2x}{dx}$

$= \frac{-1}{1+(2x)^2} \times 2$

$= \frac{-2}{1+4x^2}$ //.

d) $y = \sin^{-1}(2x-3)$

differentiating with respect to x.

$\frac{dy}{dx} = \frac{d\sin^{-1}(2x-3)}{d(2x-3)} \times d(2x-3)$

$= \frac{1}{\sqrt{1-(2x-3)^2}} \times 2$

$= \frac{2}{\sqrt{1-(2x-3)^2}}$ //.

e) $y = \sec^{-1} x^3$

differentiating with respect to x.

$$\frac{dy}{dx} = \frac{d \sec^{-1} x^3}{dx}$$

$$= \frac{d \sec^{-1} x^3}{dx^3} \times \frac{dx^3}{dx}$$

$$= \frac{1}{\cancel{x^3} \sqrt{x^6 - 1}} \times 3\cancel{x^2}$$

$$= \frac{3}{x \sqrt{x^6 - 1}}$$ #.

f) $y = \tan^{-1}(ax+b)$

differentiating with respect to x.

$$\frac{dy}{dx} = \frac{d \tan^{-1}(ax+b)}{dx}$$

$$= \frac{d \tan^{-1}(ax+b)}{d(ax+b)} \times \frac{d(ax+b)}{dx}$$

$$= \frac{1}{1 + (ax+b)^2} \times a$$

$$= \frac{a}{1+(ax+b)^2}$$

g) $y = \sec^{-1}(\tan x)$

differentiating with respect to x.

$$\frac{dy}{dx} = \frac{d \sec^{-1}(\tan x)}{dx}$$

$$= \frac{d \sec^{-1}(\tan x)}{d(\tan x)} \times \frac{d(\tan x)}{dx}$$

$$= \frac{1}{\tan x \sqrt{\tan^2 x - 1}} \times \sec^2 x$$

$$= \frac{\sec^2 x}{\tan x \sqrt{\tan^2 x - 1}}$$ #.

h) $y = \tan^{-1}\left(\dfrac{\sin x}{1+\cos x}\right)$

differentiating with respect to x.

$$\frac{dy}{dx} = \frac{d \tan^{-1}\left(\dfrac{\sin x}{1+\cos x}\right)}{dx}$$

$$= \frac{d \tan^{-1}\left(\dfrac{\sin x}{1+\cos x}\right)}{d\left(\dfrac{\sin x}{1+\cos x}\right)} \times \frac{d\left(\dfrac{\sin x}{1+\cos x}\right)}{dx}$$

$$= \frac{1}{1+\left(\dfrac{\sin x}{1+\cos x}\right)^2} \times \frac{(1+\cos x)\dfrac{d\sin x}{dx} - \sin x \cdot \dfrac{d(1+\cos x)}{dx}}{(1+\cos x)^2}$$

$$= \frac{1}{1+\dfrac{\sin^2 x}{1+2\cos x+\cos^2 x}} \times \frac{(1+\cos x)\cos x - \sin x \cdot x - \sin x}{(1+\cos x)^2}$$

$$= \frac{1}{\dfrac{1+2\cos x+\cos^2 x+\sin^2 x}{1+2\cos x+\cos^2 x}} \times \frac{\cos x+\cos^2 x+\sin^2 x}{(1+\cos x)^2}$$

$$= \frac{\cancel{(1+\cos x)^2}}{1+2\cos x+1} \times \frac{\cos x+1}{\cancel{(1+\cos x)^2}}$$

h) $y = \tan^{-1}\left(\dfrac{\sin x}{1+\cos x}\right)$

$y = \tan^{-1}\left(\dfrac{\cancel{2}\sin x/2 \cdot \cancel{\cos x/2}}{\cancel{2}\cos^{\cancel{2}} x/2}\right)$

$y = \tan^{-1}\left(\tan \dfrac{x}{2}\right)$

$y = \dfrac{x}{2}$

differentiating with respect to x.

$$\frac{dy}{dx} = \frac{d\left(\frac{x}{2}\right)}{dx}$$

$$= \frac{1}{2}$$

Ans.

③ Find the differential coefficients of following with respect to x.

a) $y = \sin^{-1}\left(\sqrt{1-x^2}\right)$

let, $x = \cos\theta \Rightarrow \theta = \cos^{-1}x$.

$y = \sin^{-1}\left(\sqrt{1-\cos^2\theta}\right)$

$y = \sin^{-1}\left(\sqrt{\sin^2\theta}\right)$

$y = \sin^{-1}(\sin\theta)$

$y = \theta$

$y = \cos^{-1}x$

differentiating with respect to x.

$$\frac{dy}{dx} = \frac{d\cos^{-1}x}{dx}$$

$$= \frac{-1}{\sqrt{1-x^2}}$$

Ans.

b) $y = \sin^{-1}(3x - 4x^3)$

let $x = \sin\theta \Rightarrow \theta = \sin^{-1}x$

$y = \sin^{-1}(3\sin\theta - 4\sin^3\theta)$

$y = \sin^{-1}(\sin 3\theta)$

$y = 3\theta$

$y = 3\sin^{-1}x$

differentiating with respect to x.

$$\frac{dy}{dx} = \frac{3\,d\left[\sin^{-1}x\right]}{dx}$$

$= 3 \times \dfrac{1}{\sqrt{1-x^2}}$

$= \dfrac{3}{\sqrt{1-x^2}}$ Ans.

c) $y = \cos^{-1}(3x - 4x^3)$

let $x = \sin\theta \Rightarrow \theta = \sin^{-1}x$

$y = \cos^{-1}(3\sin\theta - 4\sin^3\theta)$

$y = \cos^{-1}(\sin 3\theta)$

$y = \cos^{-1}\left(\cos\left(\dfrac{\pi}{2} - 3\theta\right)\right)$

$y = \dfrac{\pi}{2} - 3\theta$

$y = \dfrac{\pi}{2} - 3\sin^{-1}x$

differentiating with respect to x.

$\dfrac{dy}{dx} = \dfrac{d\left[\dfrac{\pi}{2} - 3\sin^{-1}x\right]}{dx}$

$= 0 - \dfrac{3\,d\sin^{-1}x}{dx}$

$= -3 \times \dfrac{1}{\sqrt{1-x^2}}$

$= \dfrac{-3}{\sqrt{1-x^2}}$ Ans.

d) $y = \cos^{-1}\left(\frac{2x}{1+x^2}\right)$

let $x = \tan\theta \Rightarrow \theta = \tan^{-1}x$

$y = \cos^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right)$

$= \cos^{-1}(\sin 2\theta)$

$= \cos^{-1}\left[\cancel{\sin}\ \cos\left(\frac{\pi}{2} - 2\theta\right)\right]$

$= \cos^{-1}\cos(\pi/2 - 2\theta)$

$y = \pi/2 - 2\theta \Rightarrow \pi/2 - 2\tan^{-1}x$

differentiating with respect to x.

$\frac{dy}{dx} = \frac{d[\pi/2 - 2\tan^{-1}x]}{dx}$

$= \frac{d\pi/2}{x} - 2\frac{d\tan^{-1}x}{dx}$

$= 0 - 2 \times \frac{1}{x^2+1}$

$= \frac{-2}{x^2+1}$

e) $y = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$

let $x = \tan\theta \Rightarrow \theta = \tan^{-1}x$.

$y = \tan^{-1}\left(\frac{2\tan\theta}{1-\tan^2\theta}\right)$

$y = \tan^{-1}\tan 2\theta$

$y = 2\theta$

$y = 2\tan^{-1}x$

differentiating with respect to x.

$$\frac{dy}{dx} = \frac{d\,[2\tan^{-1}x]}{dx}$$

$$= 2\,\frac{d\tan^{-1}x}{dx}$$

$$= 2 \times \frac{1}{x^2+1}$$

$$= \frac{2}{x^2+1}$$ //

f) $y = \sin^{-1}(2x\sqrt{1-x^2})$

let $x = \sin\theta \Rightarrow \theta = \sin^{-1}x$

$$y = \sin^{-1}(2\sin\theta\sqrt{1-\sin^2\theta})$$

$$= \sin^{-1}(2\sin\theta \cdot \cos\theta)$$

$$= \sin^{-1}(\sin 2\theta)$$

$$y = 2\theta \Rightarrow y = 2\sin^{-1}x$$

differentiating with respect to x.

$$\frac{dy}{dx} = \frac{d\,[2(\sin^{-1}x]}{dx}$$

$$= 2\,\frac{d\sin^{-1}x}{dx}$$

$$= 2 \times \frac{1}{\sqrt{1-x^2}}$$

$$= \frac{2}{\sqrt{1-x^2}}$$ //

9) $y = \tan^{-1} \frac{\sqrt{1+x^2} - 1}{x}$

let, $x = \tan\theta \Rightarrow \theta = \tan^{-1}x$.

$y = \tan^{-1} \frac{\sqrt{1-\tan^2\theta} - 1}{\tan\theta}$

$= \tan^{-1}\left(\frac{\sec\theta - 1}{\tan\theta}\right)$

$= \tan^{-1}\left(\frac{1-\cos\theta}{\sin\theta}\right)$

$= \tan^{-1}\left(\frac{\cancel{2}\sin^{\cancel{2}}\theta/2}{\cancel{2}\cancel{\sin\theta/2}.\cos\theta/2}\right)$

$= \tan^{-1}(\tan\theta/2)$

$y = \frac{\theta}{2} \Rightarrow y = \frac{\tan^{-1}x}{2}$

differentiating with respect to x.

differentiating with respect to x.

$\frac{dy}{dx} = \frac{d\left(\frac{\tan^{-1}x}{2}\right)}{dx}$

$= \frac{1}{2} \times \frac{d\tan^{-1}x}{dx}$

$= \frac{1}{2} \times \frac{1}{x^2+1}$

$= \frac{1}{2(x^2+1)}$ Ans.